

Local Labor Market and Land Use Effects from Deforestation Caused by an Invasive Species

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Abstract

The spotted lanternfly (SLF) is an invasive species that first appeared in the United States in 2014 and is rapidly spreading across the country. The SLF feeds on the sap from many economically important plants including grapevine, apple, and maple. Using a staggered difference-in-differences model, this paper produces the first causal estimates of the labor market effects induced by the spread of the SLF and also uncovers the mechanisms driving this result. I find that average wages decrease, the unemployment rate decreases, and the number of jobs increases in counties with established colonies of SLF. These new jobs come from lower paying sectors, specifically from construction and from low-paying urban amenities jobs. The decrease in average wages is driven largely by an industry mix effect, although productivity effects likely also contribute to the observed decrease in wages. To explain these economic outcomes, I show that the SLF causes a reduction in forested land and an increase in developed land. These land use changes appear to drive the increase in urban amenities and thus the effects on local economies. Interestingly, these communities appear to view the land use changes and increase in urban amenities as a net amenity gain as I show that rents increase using a county-level house price index.

Keywords: Deforestation; Land Use; Local Economic Development; Invasive Species; Spotted Lanternfly

JEL Codes: R11; R14; R23; Q23; Q57; J21

1 Introduction

First introduced to the United States in Pennsylvania in 2014, the spotted lanternfly (SLF) is an invasive species that feeds off of sap-producing trees and crops (De Bona et al., 2023). Many of these plants are of significant economic and ecological importance, with the grape/wine sector being particularly affected by the SLF (Harner et al., 2025). In addition to grapevine, the SLF is a known pest for tree species including many types of maple, oak, birch, beech, locust, poplar, apple, peach, and cherry, among others (Dara et al., 2015). To mitigate these impacts, some initial efforts have been implemented to try to stop or slow the spread of the SLF (Urban et al., 2021). But despite these efforts, the SLF continues to spread. From an economics perspective, this continued spread could be due in part to suboptimal investment in SLF mitigation strategies.

One reason for suboptimal investment in addressing invasive pests like the SLF is an incomplete understanding of the total damages associated with the invasive species (Crystal-Ornelas and Lockwood, 2020). As Jones (2020b) points out, there are very few economic studies on the economic costs of invasive species in general. Additionally, many methods of estimating the costs of invasive pests rely on non-market valuation techniques and therefore are costly in their own right (Hanley and Roberts, 2019). However, market-level effects attributable to invasive species can be estimated *ex post facto*.

Examples of such studies are sparse in the invasive species literature in general and non-existent for the SLF specifically. The core of the economics literature on local labor market-level effects caused by invasive species focuses on the emerald ash borer (EAB) and links EAB-induced deforestation to reductions in wages, health outcomes, and overall well-being (Jones, 2020a,b; Jones and McDermott, 2018; Jones, 2017). For example, Jones (2020b) linked reductions in wages to changes in the health of workers (a productivity effect) and the labor supply brought on by increased air pollution caused by the EAB-induced deforestation. However, it is not clear to what extent these channels contribute to reduced wages and other unwanted outcomes in local labor

markets, or if there are other channels through which invasive species affect local labor markets.

From a local economic development perspective, deforestation caused by an invasive pest can have many impacts. Deforestation - or even the risk of deforestation - can induce changes in land use. To see this, consider a scenario in which you are the owner of a timber stand. If an invasive species such as the SLF spreads to your location, the net present value of your timber decreases due to the risk of timber die-off due to the SLF (Macpherson et al., 2018). Consequentially, the best use of your land may no longer be forestry. Instead, you may find that developing your land may be the most valuable land use. The key to this paper is that developing this land implies more than just increased air pollution due to invasive species-induced deforestation. Developing previously forested land directly reduces the provision of natural amenities and can mean more businesses, more jobs, and, critically, a change in the local industry mix (Roback, 1982). Thus, reductions in average wages in a given location after the introduction of an invasive species could also be a consequence of an industry mix effect driven by the creation of low paying jobs in what I call the “urban amenities” sector (i.e., jobs in entertainment, accommodation, and food services). This industry mix effect represents an additional pathway through which land use changes brought on by invasive pests like the SLF can affect local economies.

This paper contributes to the literature in three ways. First, I provide insight into the extent to which each of these pathways (industry mix and productivity) affects wages. I do this by first estimating the effect of the establishment of SLF colonies on critical indicators of local economies including wages, household income, the unemployment rate, labor force participation, the number of people employed, the number of businesses, and house price indices at the county level using a staggered difference-in-differences model. I find that average wages decrease by 1-2% (very similar in magnitude to the wage effect induced by the EAB estimated by Jones (2020b)), the unemployment rate decreases by half a percentage point, the number of employed workers increases by about 3%, labor force participation remains unchanged, and house prices (measured by the Zillow house price index) increase by about 4%.

To explain these results, I run two additional sets of models designed to understand the mechanism driving these effects. First, to understand the potential productivity and industry mix effects, I estimate the effect of the SLF on employment across different sectors. Here I find that the employment gains were driven by increases in two sectors: construction and urban amenities. Next, I estimate models of changes in land cover. I find that the SLF does indeed induce deforestation. I also find evidence that some of the deforested land is developed. This is consistent with the increased employment in the construction and urban amenities sectors. Since these are relatively low-paying sectors, these findings support the hypothesis that an industry mix effect does indeed contribute to the reduction in wages caused by deforestation-inducing invasive pests like the SLF.

The second contribution of this paper to the literature is to describe the relative contributions of the industry mix and productivity effects. To provide insight into the relative contribution of each of these two pathways, I decompose the main result on wages into an industry mix effect and a productivity. I find that the industry mix effect explains approximately two-thirds of the wage reduction, leaving only the remaining third of the wage reduction attributable to a productivity effect. Back of the envelope calculations suggest that (as of 2023) the SLF cost workers \$1.55 billion annually, though about \$1 billion of this is attributable to the industry mix effect, leaving only about \$500 million of this to be explained by a productivity effect.

The third and final contribution of this paper is that it provides the first causal evidence of SLF-induced deforestation. Importantly, while I do find evidence of deforestation caused by the SLF, I cannot determine if this deforestation is caused directly or indirectly by the SLF. It could be the case that the majority of this deforestation is caused by profit-maximizing land use decisions born by the introduction of the SLF and resulting in land managers deciding to harvest timber and develop land.

2 Background on Invasive Species

Invasive species like the SLF must cause some sort of ecological disruption by definition; according to the US National Parks Service, species introduced by humans to a new ecosystem that do not cause harm to the “environment, economy, or human, animal, or plant health” are referred to as non-native species rather than invasive species (US National Park Service, 2020). An important economic consequence of this decision is that marginal damages from non-native species are zero by definition. This means that investing in controlling the spread of an introduced species can only improve social welfare if the species is invasive.

Understanding the costs associated with invasive species is not straightforward. Environmental and ecological impacts from invasive species can be economically significant but are often difficult to establish. The SLF, for instance, feeds on grapevine and thus affects everything further along the grape supply chain, including grape, grape juice, raisin, and wine production (Huron et al., 2022). The blue catfish, a fish species native to North America that was introduced to new ecosystems in the eastern US for the sake of increasing recreational opportunities has been shown to feed on the economically vital blue crab in the Chesapeake Bay, consuming in excess of 400 metric tons per year (Hilling et al., 2023, 2025). From an ecological perspective, invasive species have been linked to an increased risk of extinction for more than 40% of all species classified as “threatened” or “endangered” (Pimentel et al., 2005).

Controlling the population of invasive species is both difficult and costly. For one, human activity is both directly and indirectly for the spread of invasive species. In addition to the deliberate introduction of invasive species like the blue catfish (Hilling et al., 2025), silver and bighead carp (Brown et al., 2025), and wild pigs (Bevins et al., 2014), for recreational purposes, daily human activity has been linked to the spread of already present invasive species in the form of trade and land use patterns (Hulme, 2009; Richards et al., 2014). The spread of the SLF has also been linked to typical human activities, most notably transportation (Ladin et al., 2023).

Stopping the spread of invasive species is also difficult because successful eradication or containment of an invasive species requires large-scale coordination across both the spatial and temporal dimensions (Epanchin-Niell and Wilen, 2012). As Epanchin-Niell and Wilen (2015) showed in theory, cooperative management of invasive species by individuals across space leads to better outcomes compared to a non-cooperative approach, with the overall level of success in controlling the invasive species dependent upon the level of cooperation. Further complicating the optimal control of invasive species, monitoring the presence of an invasive species is costly, leading to uncertainty in our understanding of the true distribution of an invasive species at any point in time and affecting optimal control strategy (Chalak et al., 2017; Fan et al., 2020).

From an economics perspective, the human impacts of invasive species have begun to garner significant attention from researchers. In particular the EAB, an invasive insect similar to the SLF, has been linked to an array of health and economic impacts on people. The EAB targets ash trees, and has caused substantial deforestation across much of the midwestern US (Sun et al., 2024). Given the similarity between the behavior of the EAB and the SLF, this makes results on human health and productivity attributable to the EAB an excellent basis for forming expectations about the impacts of the SLF as it continues to spread across the US.

Beginning with impacts on human health, the earliest work linking the EAB to poorer health outcomes dates back to Donovan et al. (2013) who used a two-way fixed effects model to estimate the effect of the presence of the EAB on cardiovascular mortality rates. They found that the arrival of the EAB in a county led to an additional 16.7 cardiovascular deaths per 100,000 people. A related study on cardiovascular disease in women found that the EAB was associated with a 25% increase in cardiovascular disease in general (Donovan et al., 2015). The authors discuss how factors stemming from EAB-induced deforestation such as increased air pollution, decreased exercise, and increased stress are potential mechanisms for this effect. Jones and McDermott (2018) directly explored the air pollution mechanism by using the presence of the EAB as an instrument for air pollution in an instrumental variables model linking pollution exposure to cardiovascular

and respiratory mortality. They found that greater exposure to air pollution does indeed increase rates of cardiovascular and respiratory mortality, and were even able to show that the increase in air pollution directly attributable to between 6 and 32 additional cardiovascular deaths per 100,000 people per year and between 2 and 4 additional respiratory deaths per 100,000 people per year.

Pesticides are a common form of managing invasive pests like the SLF or EAB. However, pesticides are known to cause many severe health problems, including causing many forms of cancer (Cavalier et al., 2023). Using the spotted wing drosophila (another invasive insect that affects fruit trees) as an instrument for pesticide application, Jones (2020a) found that increased pesticide exposure caused increases in premature births and in babies born at a low birthweight. This further demonstrates the far-reaching human health effects caused by invasive species.

In addition to air pollution and pesticide exposure, invasive species can negatively affect human health through changes in how people allocate their time between labor and leisure activities. Jones (2016) found that people living in counties where the EAB was present shifted their time away from leisure activities, instead spending more time participating in the labor market after the EAB spread to their home county. However, no specific mechanism for this effect was provided. Possibly due to this decrease in leisure time, the EAB has also been shown to cause a decrease in physical activity and an accompanying 2.5 percentage point increase in obesity rates (Jones, 2023). Coinciding with these effects, Jones (2017) found that the people living where the EAB had spread also experienced an overall decrease in their perceived life satisfaction.

Contributing to this decline in quality of life from a different angle, the EAB has been also been shown to negatively affect labor market outcomes for workers in affected counties. In the most similar study in the literature to my paper, Jones (2020b) used a difference-in-differences model to estimate the effect of the EAB on outcomes including wages, labor force participation, the number of employed workers, the number of retirees, and retirement benefits per retiree. They use a theoretical framework where wages are a function of environmental quality. In this context, the EAB could affect wages by reducing environmental quality through increased air pollution

arising as a consequence of deforestation. They found that the EAB decreased wages in affected counties by 1%. They found no impact on labor force participation but did find an increase in the number of retirees. They also found weak evidence (only significant at the 10% level) for a decrease in the number of employees and an increase in retirement benefits. These results are important because they collectively show that an invasive pest can affect the economic livelihood of people of people living in infested communities.

I build on this literature generally (and Jones (2020b) specifically) by using the introduction of the invasive SLF to explore the specific mechanism through which invasive pests affect local labor markets. Jones (2020b) attributed all of the labor market effects to an EAB-induced decline in environmental quality affecting (1) worker productivity via a health effect and (2) a labor supply effect. I dig deeper into the potential mechanisms by linking SLF-induced changes in land use to an industry mix effect. I show that (1) the SLF caused deforestation, (2) deforested land is replaced with developed land as the use value of land changes, (3) there is an increase in jobs in the low-paying construction and urban amenities sectors, (4) average wages decrease, (5) the number of employees increases while labor force participation holds steady, leading to a decrease in unemployment, and (6) this industry mix effect explains two-thirds of the decrease in average wages.

These results are important because understanding the full set of costs associated with an invasive species is a critical factor in determining optimal control policy. Furthermore, it is not clear how a decrease in average wages caused by an industry mix effect should be taken into consideration when accounting for the costs of an invasive species since as much as two-thirds of the decrease in average wages is a consequence of increased employment.

3 Data & Methodology

3.1 Data

The data used in this study can be grouped into three categories: spatiotemporal data on the distribution of the SLF, county-level economic data, and land use data. These data are restricted in space to the contiguous 48 US states and in time to the period from 2010 to 2023. Summary statistics for the data can be found in Table 2.

The SLF data comes from De Bona et al. (2023). This data set was created by combining many smaller data sets from government agencies and citizen science projects. This data set contains information on both individual SLF sightings and established SLF colonies. Because individual bugs can hitch rides on vehicles or in shipments, I use only established SLF colonies to define the presence of the SLF in a given county. This data set is built into the *lydemapr* package in R. The spread of the SLF is depicted in Figure 1.

The economic data consists of data on outcomes and relevant control variables. The outcomes included here are wages, household income, the unemployment rate, the labor force participation rate, the number of employed workers, the total population, the number of business establishments, the number of farms, and a house price index. The controls included here pertain to education, racial and age demographics, and the housing market. I include the percentage of the population 25 years or older with at least a college degree, racial and ethnic composition, the proportion of elderly residents, and the number of housing units in each county.

Data on average wages comes from both the Internal Revenue Service (IRS) and from the US Census Bureau County Business Patterns (CBP). These wage data measure the average wage for all workers in a given county each year. Due to data availability, the IRS wage data only covers the years from 2010 through 2022.

Data on household income comes from the American Community Survey (ACS). Specifically,

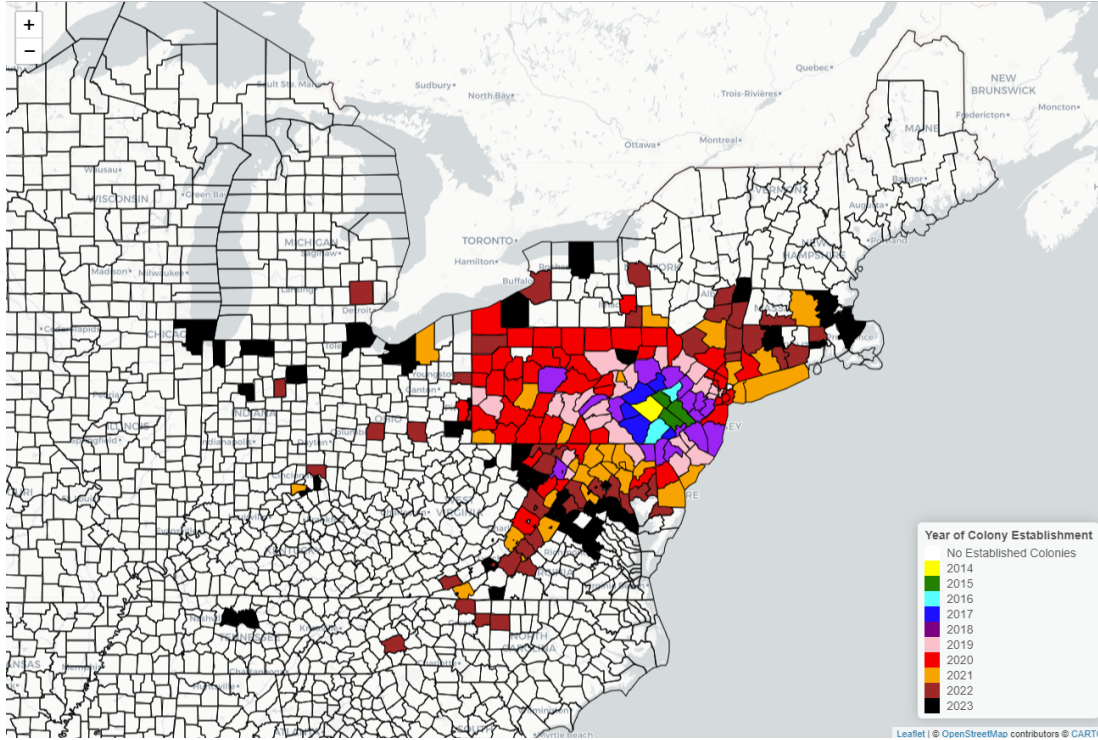


Figure 1: This figure shows the spread of the SLF across counties by year of first established colony.

I use data on median household income in each county from 2010 to 2023. Data on the unemployment rate, labor force participation measured as the number of participating individuals, and total county population also comes from the ACS. Data on the control variables also comes from the ACS.

Data on the number of employed workers and on the number of business establishments operating in counties comes from the CBP data. In addition to the number of business establishments generally, I also use data on the number of farm establishments from the IRS. I include this as an outcome because the SLF is known to be a pest to certain agricultural commodities such as apple and grapevine (Multi-Resolution Land Characteristics Consortium, 2024; Harner et al., 2025). This data is restricted to the years 2010 through 2022.

The last outcome studied here addresses the rents component of a spatial equilibrium model. To do this, I use the Zillow Home Value Index. This index is designed to estimate a reasonable

price for a median valued house in each county at a given point in time.¹

To study potential industry mix effects, I use data on county-level employment across the thirteen industry sectors documented the ACS. This data spans 2010 through 2023, and these thirteen sectors are listed in Table 1. These industry sectors are based aggregations of smaller industries defined in the North American Industry Classification System (NAICS). Detailed information on how these thirteen sectors are defined can be found on the ACS website.² For the purposes of this study, I refer to the “Arts, entertainment, and recreation, and accommodation and food services” sector as the “urban amenities” sector.

Table 1: Industry Sectors in the ACS

Industry Sector
1: Agriculture, forestry, fishing and hunting, and mining
2: Construction
3: Manufacturing
4: Wholesale trade
5: Retail trade
6: Transportation and warehousing, and utilities
7: Information
8: Finance and insurance, and real estate and rental and leasing
9: Professional, scientific, and management, and administrative and waste management services
10: Educational services, and health care and social assistance
11: Arts, entertainment, and recreation, and accommodation and food services
12: Other services, except public administration
13: Public administration

The land use data comes from two sources. First, I use the National Land Cover Database (NLCD) which provides data on the proportion of land allocated to each land cover type (Multi-Resolution Land Characteristics Consortium, 2024). I use eight land cover categories that collectively account for 100% of the surface area of each county. These eight categories are forested land, developed land, agriculture, shrubland, grassland, wetlands, water, and barren land. As I will explain in the next subsection where I outline the methodologies used in this paper, this data will be fed into a differenced model. I will compare the change in surface area allocated to each of these

¹<https://www.zillow.com/research/data/>

²<https://www.census.gov/programs-surveys/acs.html>

eight land cover categories in each county between 2011 and 2021.

The second source for land use data comes from Sanda and Lindeman (2025). This is a data set comprised of changes in forest cover at the county level between 2010 and 2022 for counties in the eastern US. This data set combines data from the NLCD, the LANDFIRE Existing Vegetation data, and the Landscape Change Monitoring System to create enhanced estimates of deforestation (Sanda and Lindeman, 2025). This data set is used to ensure the robustness of the land use dynamics results using the NLCD data set.

Table 2: Summary Statistics

Category	Variable	Mean	Std. Dev.
Ecological	Spotted Lanternfly	0.017	0.128
	ln Wages (IRS; \$)	10.704	0.236
Economics	ln Wages (CBP; \$)	10.541	0.251
	ln Median Household Income (ACS; \$)	10.805	0.276
	Unemployment Rate (ACS)	0.067	0.035
	ln Labor Force Size (ACS)	9.516	1.513
	ln Population (ACS)	10.281	1.475
	ln Employees (CBP)	8.875	1.784
	ln Establishments (CBP)	6.450	1.505
	ln Farms (IRS)	6.006	1.082
	ln House Price Index (Zillow; \$)	11.977	0.500
Land Use	Forests (% of Surface Area)	0.293	0.252
	Developed (% of Surface Area)	0.092	0.118
	Agriculture (% of Surface Area)	0.316	0.268
	Shrublands (% of Surface Area)	0.084	0.181
	Grasslands (% of Surface Area)	0.093	0.168
	Wetlands (% of Surface Area)	0.070	0.105
	Water (% of Surface Area)	0.049	0.115
	Barren (% of Surface Area)	0.004	0.014
Deforestation	Deforestation (% Forest Loss)	0.002	0.003
	Normalized Deforestation (% Forest Cover)	0.005	0.009
	Forest Cover (%)	0.472	0.271

3.2 Methodology

To estimate the effects that the invasive SLF has on the local labor market outcomes outlined in the previous subsection, I use the staggered difference-in-differences model developed by Callaway and Sant’Anna (2021b). The ability to incorporate staggered treatment timing here is critical because the SLF can spread not only by expanding its territory to adjacent counties via population growth, but also by interstate commerce (Ladin et al., 2023; Strömbom et al., 2024). The latter means allows for multiple, distinct sources from which the SLF can spread.

The staggered difference-in-differences method is an improvement on the traditional difference-in-differences model primarily due to the fact that it addresses the issue of bias arising from assigning negative weights to units when computing the average treatment effect on treated units (ATT) when there are multiple time periods and when the timing of treatment varies across treated units (Goodman-Bacon, 2021).

The Callaway and Sant’Anna (2021b) staggered difference-in-differences model used here can estimate an effect ATT using either one of two possible control groups: one comprised of the not yet treated units (i.e., those counties to which the SLF eventually spread, but has not yet spread to by a given year), and one comprised solely of the never treated units (i.e., those that never have an established SLF colony by the end of this study). I use the never treated control group since there is no potential for treatment contamination in the control counties. However, because these never treated units could be fundamentally different from the treated units, I also employ coarsened exact matching on the control variables. The coarsened exact matching process used here follows Cary and Orteiz (2026) and matches treated counties to controls that have population levels within 5% of the treated county, median household incomes within 5% of the treated county, unemployment rates within half of a percentage point of the treated county, and college attainment rates within 5% of the treated county.

The specific difference-in-differences model with staggered treatment timing used here is

specified in Equation 1. In this model, $Y_{i,s,t}$ is one of the economic outcomes of interest described above for county i in state s for year t , $SLF_{i,t}$ is an indicator variable equal to one if SLF colonies have been established in county i at year t , $X_{i,t}$ is a vector of control variables at the county-year level, $\delta_{s \times t}$ are state-by-year fixed effects, and $\epsilon_{i,s,t}$ is the model residual. State-by-year fixed effects were accounted for in this model by first demeaning outcomes at the state-year level and then using the demeaned outcomes in the Callaway and Sant’Anna (2021b) regression using the *did* library in R developed by Callaway and Sant’Anna (2021a).

$$Y_{i,s,t} = \theta SLF_{i,t} + X_{i,t}\beta + \delta_{s \times t} + \epsilon_{i,s,t} \quad (1)$$

To ensure that a potential claim of causality can be made here, I test for parallel pre-trends. Verifying parallel pre-trends for the treated and control groups is critical for inference here since it justifies the claim that the counties with SLF would have seen similar outcomes to the untreated counties had the SLF never been introduced. Parallel pre-trends are verified for all outcomes studied using the model specified by Equation 1. Event study plots verifying the validity of the parallel pre-trends assumption for the difference-in-differences model for key outcomes that yields statistically significant results are presented in Figure 2.

In addition to the local economic outcomes above, I estimate the effect of SLF on land use using the differenced model specified in Equation 2. To do this, I use the deforestation data set from Sanda and Lindeman (2025) and general land use data from the National Land Cover Database (Multi-Resolution Land Characteristics Consortium, 2024).

In this model, the outcomes $\Delta Y_{i,s}$ are county level changes in land cover between two periods of time, one before the SLF arrived in the US and one after. The land cover data from the NLCD compares the distribution of land cover across eight categories (forested land; developed land; agriculture; shrubland; grassland; wetlands; water; barren land) in 2011 and 2021. The deforestation data set from Sanda and Lindeman (2025) consists of changes in forested land between 2010 and 2022. The variable SLF_i is indicator variable equal to one if a SLF colony was

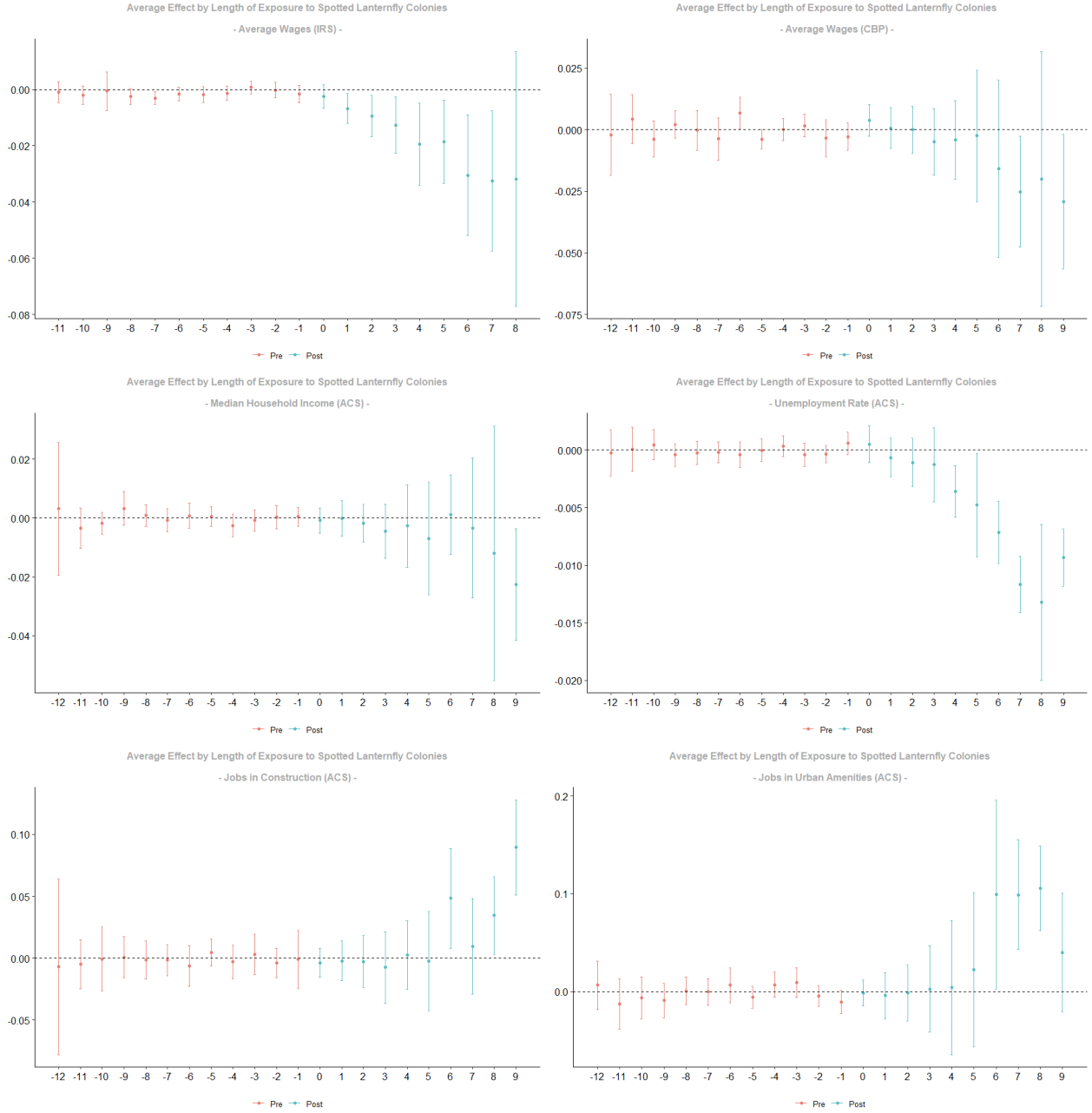


Figure 2: Event Study Plots from the Callaway-Sant'Anna Model

established in county i by the end of the study. I also run a version of this model where the variable SLF_i is an intensity, measured by the number of years after the first SLF colony was established in county i . The rest of the model includes changes in county-level controls during this period are contained in X_i , a state fixed effect δ_s to account for the development of land use policy variation

across states, and $\varepsilon_{i,s}$ is the residual. Since this is a model of differences between two periods, I rely on the same coarsened exact matching process used with Equation 1 to ensure that the control group is indeed representative of the treated group.

$$\Delta Y_{i,s} = \theta SLF_i + \Delta X_i \beta + \delta_s + \varepsilon_{i,s} \quad (2)$$

4 Results and Discussion

4.1 *County Level Economic Outcomes*

I begin by presenting and discussing the main results on county-level economic conditions using the staggered difference-in-differences model specified in Equation 1. These outcomes include average wages, median household income, the unemployment rate, the number of people participating in the labor force in any capacity, the number of employed people, total population, the number of business establishments, the number of farm establishments, and the house price index. Results using the full sample of all counties in the contiguous US are presented in Table 3, and results using the coarsened exact matching sample are presented in Table 4. While all results presented in these tables use the never treated units as the control group, all results are robust to using the not yet treated units as the control group.

Beginning with the wage effect, I find that average wages decreased by anywhere from 1-2% after the establishment of SLF colonies. This effect is consistent across both samples (full sample and coarsened exact matching), but differs in magnitude somewhat across data sets. The estimated wage effect using the IRS data is larger (1.8-2.0%) than the wage effect estimated using the CBP data (1.0-1.1%). However, these effects are similar to that estimated by Jones (2020b) for the invasive emerald ash borer, especially the results using the CBP data.

Moving on to median household income, I find at most weak evidence that household incomes

declined very modestly, by well less than 1%. This effect is statistically insignificant for the full sample and only statistically significant at the 10% level for the matched sample. Thus, I do not claim any reduction in household incomes despite a clear decrease in average wages.

The apparent contradiction emerging from decreased average wages with no change in household income can be explained in part by the very next result. The unemployment rate in counties with SLF colonies decreased substantially after the colonies were established, decreasing by half of a percentage point. Combined with no change in the size of the labor force or local population, this suggests that there was an increase in jobs attributable to the SLF. And this is perfectly consistent with the result for the number of employees where the number of employed people increased by approximately 3% after the introduction of the SLF. This set of results tells a clear story of increased employment opportunities attributable to the SLF, albeit in the form of low-paying jobs. All of these results are also consistent across both samples.

From a business perspective, I find no change in the number of business establishments. This result even extends to farm operations. This null result could be due to either the SLF bearing no meaningful effect on business entrance and exit, or the SLF causing some businesses to close while also creating new business opportunities, thereby causing no net change in the number of business. This latter possibility is very unlikely for farms, but may be applicable to non-farm sectors.

Finally, I find a substantial increase in rents of around 4% that is consistent across both samples. This result can be explained in the context of a spatial equilibrium framework by an increase in net amenities. While this may seem counterintuitive given that the SLF is an invasive species linked to deforestation and agricultural damages, if the deforested land is developed, thereby allowing for an expansion of urban amenities (and corresponding low-paying job opportunities), residents could view this as a net amenity gain. Assuming that marginal desires for increased urban amenities are higher among rural individuals and that marginal amenity values for these same rural individuals are relatively low, the rural bias of the SLF could feasibly induce land use changes that are consistent with perceived net amenities. This possibility is strengthened by the fact that I find no evidence of

population decline and since out-migration is a known consequence of reduced natural amenities (Chi and Marcouiller, 2013; Gao et al., 2023; Cary and Orteiz, 2026).

An alternative (or perhaps jointly applicable) possible explanation for this result is that deliberate harvesting of timber to prevent the spread of the SLF could be viewed as an ecosystem service. In this scenario, the previous literature suggests that house prices should increase (Kim et al., 2016). However, it is also possible that a reduction in natural amenities (e.g., forest cover) could decrease house prices; after all, the previous literature on natural amenities and house prices supports this hypothesis (Gibbons et al., 2014; Wolf et al., 2022). Further analysis of the impact of the SLF on house prices using data on real estate transactions rather than a house price index would shed more light on precisely why rents increase at the county-level. However, since the house price index is a county-level measure, the increase in net amenities argument appears the most likely driver of the increase in rents observed here. I will provide substantial evidence supporting this argument in Subsection 4.3.

In the following subsections, I build on these results by diving deeper into the mechanisms underpinning these local economic effects caused by the SLF. I will show that low wage sectors are indeed the sectors experiencing growth after the introduction of the SLF, and that the SLF does indeed induce land use changes consistent with the story as told thus far.

4.2 Heterogeneous Job Growth Across Sectors

Next I present and discuss the results on how the SLF affected employment across industry sectors using the same staggered difference-in-differences model used for the previous set of results. I present results using the full sample of counties in the contiguous US (Table 5) and results using the coarsened exact matching sample (Table 6). All results presented in these tables use the never treated units as the control group, but all results are robust to using the not yet treated units as the control group.

Table 3: Full Sample Regression Results - Labor Market Outcomes

Outcome	Estimate	Std. Error	Observations
ln Wages (IRS)	-0.018***	0.004	40,409
ln Wages (CBP)	-0.010***	0.004	43,518
ln Household Income	-0.005	0.004	43,518
Unemployment Rate	-0.005***	0.001	43,518
ln Labor Force Participation	0.001	0.004	43,518
ln Employees	0.037***	0.008	43,518
ln Population	0.005	0.005	43,518
ln Establishments	-0.003	0.006	43,518
ln Farms	-0.001	0.008	40,409
ln House Price Index (ZHVI)	0.045***	0.006	39,412

This table contains the estimated effects of SLF on a series of labor market outcomes using the staggered difference-in-differences model from Equation 1 with the full sample of all 3,108 counties in the contiguous US. The results presented here are from models using the never treated units as the control group. Standard errors are clustered at the USDA ERS commuting zone level. Statistical significance is denoted by * for $p < .10$, ** for $p < .05$, and *** for $p < .01$.

Of the thirteen industry sectors here, only two yielded statistically significant results across both sample. These two sectors were the construction sector and what I have called the urban amenities sector (arts, entertainment, and recreation, and accommodation and food services).

Beginning with the construction sector, I find a 1.3-1.7% increase in construction jobs. Turning to the other sector with a statistically significant result, urban amenities, I find a 3.6-3.7% increase in employment after the establishment of the SLF. These two results are consistent with a scenario in which forested land owners respond to the SLF by harvesting their timber early and developing the land.

To better understand this, consider the effect of the SLF on the value of a timber stand. If the SLF poses a risk to timber stands, land owners must either risk losing some of their timber or pay additional costs to address this risk (e.g., pesticide application or other control/eradication methods). Either way, the net present value (NPV) of the forested land decreases, and equally importantly, the optimal harvest date for the timber becomes earlier (Macpherson et al., 2018).

Table 4: Matched Sample Regression Results - Labor Market Outcomes

Outcome	Estimate	Std. Error	Observations
ln Wages (IRS)	-0.020***	0.005	36,596
ln Wages (CBP)	-0.011***	0.004	39,412
ln Household Income	-0.006*	0.004	39,412
Unemployment Rate	-0.005***	0.001	39,412
ln Labor Force Participation	-0.001	0.005	39,412
ln Employees	0.033***	0.008	39,412
ln Population	0.003	0.005	39,412
ln Establishments	-0.003	0.006	39,412
ln Farms	0.006	0.011	36,596
ln House Price Index (ZHVI)	0.043***	0.006	39,412

This table contains the estimated effects of SLF on a series of labor market outcomes using the staggered difference-in-differences model from Equation 1 with the restricted sample of counties obtained by coarsened exact matching. The results presented here are from models using the never treated units as the control group. Standard errors are clustered at the USDA ERS commuting zone level. Statistical significance is denoted by * for $p < .10$, ** for $p < .05$, and *** for $p < .01$.

This implies two critical economic facts.

First, the earlier timber harvest date means that the land will be cleared sooner than otherwise would have been the case, allowing for the land to be developed sooner. Second, the decreased value of the land when forested means that for some parcels, developing the land will become the profit maximizing land use. Thus, an expansion of urban amenities and an increase in employment in construction and urban amenities after the establishment of the SLF has a clear economic motivation.

Given this clear economic story of how an invasive species like the SLF can affect local economies through changes in land use, it only remains to be shown that these changes in land use do indeed follow the establishment of the SLF. This is precisely what I do in the next subsection.

Table 5: Full Sample Regression Results - Jobs by Sector

Outcome	Estimate	Std. Error	Observations
In Agriculture, forestry, fishing and hunting, and mining	0.000	0.033	43,518
In Construction	0.017**	0.007	43,518
In Manufacturing	0.007	0.010	43,518
In Wholesale trade	0.024	0.019	43,518
In Retail trade	0.012	0.009	43,518
In Transportation and warehousing, and utilities	0.051	0.039	43,518
In Information	0.003	0.016	43,518
In Finance and insurance, and real estate and rental and leasing	-0.039	0.027	43,518
In Professional, scientific, and management, and administrative and waste management services	-0.002	0.009	43,518
In Educational services, and health care and social assistance	0.007	0.005	43,518
In Arts, entertainment, and recreation, and accommodation and food services	0.036***	0.011	43,518
In Other services, except public administration	-0.003	0.010	43,518
In Public administration	-0.008	0.016	43,518

This table contains the estimated effects of SLF on the number of jobs in each sector using the staggered difference-in-differences model from Equation 1 with the full sample of all 3,108 counties in the contiguous US. The results presented here are from models using the never treated units as the control group. Standard errors are clustered at the USDA ERS commuting zone level. Statistical significance is denoted by * for $p < .10$, ** for $p < .05$, and *** for $p < .01$.

Table 6: Matched Sample Regression Results - Jobs by Sector

Outcome	Estimate	Std. Error	Observations
In Agriculture, forestry, fishing and hunting, and mining	-0.006	0.029	39,412
In Construction	0.013*	0.007	39,412
In Manufacturing	0.002	0.009	39,412
In Wholesale trade	0.023	0.019	39,412
In Retail trade	0.011	0.010	39,412
In Transportation and warehousing, and utilities	0.046	0.034	39,412
In Information	0.000	0.016	39,412
In Finance and insurance, and real estate and rental and leasing	-0.045*	0.024	39,412
In Professional, scientific, and management, and administrative and waste management services	-0.002	0.008	39,412
In Educational services, and health care and social assistance	0.006	0.005	39,412
In Arts, entertainment, and recreation, and accommodation and food services	0.037***	0.010	39,412
In Other services, except public administration	-0.005	0.011	39,412
In Public administration	-0.008	0.015	39,412

This table contains the estimated effects of SLF on the number of jobs in each sector using the staggered difference-in-differences model from Equation 1 with the restricted sample of counties obtained by coarsened exact matching. The results presented here are from models using the never treated units as the control group. Standard errors are clustered at the USDA ERS commuting zone level. Statistical significance is denoted by * for $p < .10$, ** for $p < .05$, and *** for $p < .01$.

4.3 Land Use Dynamics

Lastly, I present and discuss the results on how the SLF affected land use. These results come from using the differenced model specified in Equation 2 rather than a staggered difference-in-differences model like the previous results. These results rely on two different data sets. Results for land use changes across eight categories of surface area using the NLCD data are presented in Table 7. Results on deforestation specifically using the data from Sanda and Lindeman (2025) are presented in Table 9. Since these are differenced models using a cross section, I present results from distinct regressions using (1) a binary version of the SLF variable indicating if the SLF established a colony in the county prior to the end of the time period, and (2) the number of years that the SLF had been established in a given county by the end of the time period. Because the extent of treatment is critical to the local economic and land use effects induced by the SLF, I focus on the results using the number of years the SLF has been established in a county over the results from the binary SLF variable.

Beginning with the land use categories results in Table 7, I find strong evidence of the SLF affecting land use. Consistent with the previous literature that has already shown that the SLF impacts forests and agriculture, I find statistically significant decreases in each category of land cover, with the SLF causing an average decrease in the proportion of forested area in counties by 0.1% of total surface area for each year of infestation. This effect is smaller for agriculture, with the SLF causing a decrease in land allocated to agriculture of 0.03% of total surface area.

To motivate the local economic effects I have presented thus far, I suggested that the land attributed to forest cover prior to the establishment of the SLF is eventually developed. This is precisely what I find. Specifically, I find that for each year the SLF is present in a county, that county sees an increase in developed land equal to 0.03% of its total surface area.

In addition to forests, agriculture, and developed land, I also find that the SLF affects grasslands, shrublands, and wetlands. Specifically, the SLF increases the proportion of land allocated to

grassland and shrublands but decreases the amount of wetlands. The increase in grasslands and shrublands may be land that was previously forested but has lost its forest cover due to the SLF. Given the importance of wetlands to preserving local biodiversity, this decrease in wetlands is a critical, new externality linked to the invasive SLF that warrants further study.

So far I have only posited that the development takes place only after forested land is deforested. To better understand the true dynamics of the changes in land cover, I alter the model specified in Equation 2 by turning the SLF variable into a fixed effect that indicates the number of years that the SLF was established in each county by the end of the sample. This allows me to estimate potential nonlinear effects of the SLF on land cover over time. These results are presented in Table 8.

The results from the fixed effects model do validate the hypothesis that the SLF-induced deforestation precedes land development. Statistically significant forest loss appears for counties that have had established colonies of the SLF for at least three years, and this effect is statistically significant for all cohorts of counties with the SLF established for at least three years. Land development is weakly statistically significant (10% level) and positive for the cohort of counties that were treated for exactly four years by the end of this study, but for the cohorts treated for five and six years, the effect is strongly significant (1% level) and positive, suggesting that land development does indeed follow SLF-induced deforestation. Oddly, the maximally treated cohort had a negative and significant effect. However, this would be the smallest cohort of the counties where the SLF first appeared in the US. It may be the case that these counties are fundamentally different in their approach to controlling the SLF in some manner given that they are the source for the spread of this invasive species.

Overall, the three sets of results presented in this paper strongly support the notion that the invasive SLF directly alters land use through deforestation effects, and these land use changes impact local economies by creating low-paying jobs in the construction and urban amenities sectors. This decreases average wages, but increases employment, leading to no change in household incomes. This is direct evidence that the decrease in average wages is driven by an

industry mix effect. In the next subsection I use these results to address the possibility of a productivity effect also contributing to the decrease in average wages.

Table 7: Regression Results - Land Use (NLCD)

Outcome	SLF - Binary		SLF - Weighted		Observations
	Estimate	Std. Error	Estimate	Std. Error	
Forests	-0.001	0.001	-0.001***	0.0003	3,107
Developed Land	-0.0002	0.0003	0.0003***	0.0001	3,107
Agriculture	0.0001	0.0002	-0.0003***	0.0001	3,107
Shrublands	0.001	0.001	0.001**	0.0003	3,107
Grasslands	0.0003	0.0005	0.001***	0.0001	3,107
Wetlands	-0.0002*	0.0001	-0.0001*	0.00003	3,107
Water	0.0003***	0.0001	0.00002	0.00003	3,107
Barren	-0.00002	0.0003	0.00002	0.00004	3,107

This table contains the estimated effects of SLF (as both a binary variable and as the number of years with an infestation present) on each of eight categories of land use using data from the NLCD and a differenced model considering changes in land use between 2011 and 2021. Standard errors are clustered at the state level. Statistical significance is denoted by * for $p < .10$, ** for $p < .05$, and *** for $p < .01$.

4.4 Industry Mix or Productivity Effect?

While I have shown how an invasive species affects land use and in turn entire local economies, one key question remains: Does the observed industry mix effect explain the decrease in wages, or is there also a potential productivity effect à la Jones (2020b)? Given that similar invasive pest species have been linked to higher levels of air pollution via deforestation (Jones and McDermott, 2018), and since air pollution is known to caused a decrease in worker productivity (Chang et al., 2016), it is entirely reasonable that there would be both an industry mix effect and a productivity effect at play.

To answer this question, I use the coefficients from Table 6 to adjust average wages in each

Table 8: Fixed Effects Model Results - Land Use (NLCD)

	Developed	Forests	Agriculture	Shrublands	Grasslands	Wetlands	Water	Barren
1 Year	-0.0004*** (0.0002)	0.00004 (0.001)	0.0001 (0.0002)	0.0001 (0.001)	-0.0003 (0.0004)	-0.0003** (0.0001)	0.0004*** (0.0001)	-0.0001 (0.0002)
2 Years	-0.001 (0.0003)	0.004* (0.002)	0.001*** (0.0003)	0.002 (0.002)	0.001* (0.001)	-0.0001 (0.0001)	0.0001 (0.0001)	0.001** (0.0002)
3 Years	0.001 (0.001)	-0.003** (0.002)	-0.0004 (0.001)	0.002 (0.002)	0.001** (0.001)	-0.0001 (0.0001)	0.0001 (0.0001)	-0.0005 (0.001)
4 Years	0.0004* (0.0002)	-0.007*** (0.001)	-0.001*** (0.0002)	0.005*** (0.001)	0.003*** (0.001)	-0.0004*** (0.0001)	0.0002** (0.0001)	-0.0002 (0.0002)
5 Years	0.002*** (0.0004)	-0.004*** (0.001)	-0.003*** (0.0003)	0.004* (0.001)	0.002*** (0.001)	-0.0003* (0.0002)	0.00003 (0.0002)	0.001** (0.0002)
6 Years	0.003*** (0.0003)	-0.005*** (0.001)	-0.003*** (0.0002)	0.002** (0.001)	0.002*** (0.001)	-0.0004*** (0.0002)	0.0005*** (0.0001)	0.0005** (0.0002)
7 Years	-0.001*** (0.0003)	-0.004*** (0.001)	-0.00001 (0.0002)	0.002* (0.001)	0.0003*** (0.001)	-0.001*** (0.0001)	0.0003*** (0.0001)	0.001*** (0.0002)
Controls	✓	✓	✓	✓	✓	✓	✓	✓
State FE	✓	✓	✓	✓	✓	✓	✓	✓
Observations	3,107	3,107	3,107	3,107	3,107	3,107	3,107	3,107

This table contains the estimated effects of SLF using fixed effects for the number of years with an infestation present on land cover on each of eight categories of land use using data from the NLCD and a differenced model considering changes in land use between 2011 and 2021. Standard errors are clustered at the state level. Statistical significance is denoted by * for $p < .10$, ** for $p < .05$, and *** for $p < .01$.

sector as a “back of the envelope” exercise. Doing so, I find that average wages decrease by 0.73% due to the industry mix effect alone. Taking the 1.1% decrease in average wages from Table 4 using the CBP wage data, this industry mix effect explains 66.6% of the decrease in average wages. This means that a productivity effect could explain up to one third of the decrease in average wages attributable to the SLF. This is important because it suggests that the productivity effect argued by Jones (2020b) in the case of a similar invasive species is indeed a contributing factor to the local economic impacts induced by invasive species, and that invasive species affect workers through multiple channels. If the IRS wage effect (2% decrease) is used, the contribution of the industry mix effect on the average wage decrease falls to 36.6%. Given the similarity of the SLF and the EAB studied by Jones (2020b) and the similarity of my results using the CBP wage data with

Table 9: Regression Results - Deforestation using data from Sanda and Lindeman (2025)

	Forest Loss Pct	Normalized Forest Loss	Forest Loss Pct	Normalized Forest Loss
SLF - Binary	0.001* (0.0004)	0.005*** (0.001)		
SLF - Weighted			0.0002* (0.0001)	0.002* (0.001)
Housing Units	0.014*** (0.003)	0.035*** (0.008)	0.014*** (0.003)	0.034*** (0.008)
Forest Cover Pct	0.001** (0.001)		0.001** (0.001)	
County Demographics	✓	✓	✓	✓
State FE	✓	✓	✓	✓
Observations	2,037	2,037	2,037	2,037
Adjusted R ²	0.389	0.269	0.389	0.269

This table contains the estimated effects of SLF (as both a binary variable and as the number of years with an infestation present) on forest loss using a differenced model and data that measures forest change between 2010 and 2022 from (Sanda and Lindeman, 2025). The outcome Forest Loss Pct represents the percent of county area that experienced permanent deforestation between 2010 and 2022. The outcome Normalized Forest Loss represents the percent of forest area in 2010 that was deforested permanently as of 2022. Robust standard errors are presented in parentheses below each estimate. Statistical significance is denoted by * for $p < .10$, ** for $p < .05$, and *** for $p < .01$.

the results from Jones (2020b), I prefer the analysis using the CBP wage data. However, further research could be performed to clarify the role of these two channels in contributing to the negative wage effect brought on by invasive species like the SLF.

To illustrate the importance of differentiating between the industry mix effect and a productivity effect, I calculate the annual total wages lost due to (1) the reduction in average wages overall, and (2) the reduction in wages after removing the industry mix effect. Applying this coefficient for average wage losses on counties with SLF infestations as of 2023, this equates to over \$1.5 billion in annual lost wages. However, if I exclude the industry mix effect and only consider the productivity effect, I still find over \$500 million in lost wages each year.

While further spread of the SLF means that these figures will only grow, being able to differentiate between these two effect is critical for setting optimal control policy for deforestation-

inducing invasive species. The annual wage loss estimate from the EAB from Jones (2020b) was assumed to be due exclusively to a productivity effect. However, if we are agnostic towards industry mix effects (after all, these are new jobs), the overall economic losses due to deforestation-inducing invasive species like the SLF are much lower than previous studies have estimated. This means that optimal control policy (if industry mix effects are ignored) would be even less capable of stopping the spread. For this reason and because new invasive species continues to spread³, it is critical that agricultural and environmental economists identify other economic losses attributable to invasive species so that we can set policy that successfully addresses invasive pests like the SLF.

5 Conclusion

In this paper I have shown that the invasive SLF fundamentally alters local economies through its effects on land cover and land use. In particular I find that the SLF caused a decrease in forest cover and an increase in developed land. This increase in developed land brings an increase in low-paying jobs in the urban amenities and construction sectors. This industry mix effect causes employment to increase but causes average wages to decrease. While the magnitude of the industry mix effect explains the majority of the observed wage effect, I cannot rule out the possibility that a productivity effect also contributes to the decrease in average wages. This is important because the SLF is not the first - and almost certainly will not be the last - invasive pest that affects forest cover and agriculture in the US. Understanding the local economic impacts induced by these invasive species is critical to setting optimal policy regarding the control of such invasive species.

These results have a profound impact on optimal control policy for invasive species linked to deforestation. Given the continued spread of the SLF, species-specific control policy is clearly warranted. Since the public is not always willing to engage in the eradication of invasive species

³The screw worm, considered an invasive species in the US, has begun to infest cattle in the US: <https://www.theguardian.com/us-news/2026/jun/16/screwworm-cases-rise-outbreak-spreads>.

(Kleespies et al., 2024; Sharp et al., 2011), the increased job opportunities arising from changes in land use may ultimately add an additional barrier to controlling invasive pests. Furthermore, if we only consider the productivity effect and ignore the industry mix effect, the costs of invasive species like the SLF decrease substantially. However, it is quite unlikely that we have identified all of the economic consequences of invasive species. Identifying the full set of costs associated with invasive species is an important open problem facing economists.

This paper also contributes to the environmental science literature by providing the first causal evidence of deforestation attributable to the SLF. To be clear, I do not show that the SLF directly deforests land; as I discussed earlier in the paper, it could also be the case that land owners with forested land harvest their timber earlier than they otherwise would due to the encroachment of the SLF. Determining exactly why forest cover is lost after the SLF appears is still an important open question.

In addition to this question, many more questions about the impacts of the SLF and of invasive species more broadly still exist. For instance, I found here that the proportion of land allocated to agriculture decreases modestly in areas where the SLF spreads, but I could not answer why this is the case. This could be due to a crop-specific decline in planting area (e.g., grapevine) or it could be driven by a behavioral response among land owners since the SLF affects the value of different types of land use.

From an economics perspective, many more questions emerge. Given the deforestation linked to the SLF, how affected is the outdoor recreation sector? Is tourism affected in locations subject to this deforestation? For those rural communities reliant upon tourism for outdoor recreation, this could have critical, long-term impacts on the local economy. In addition to recreation and tourism, are there health or other well-being effects caused by the SLF and the deforestation ensuing from it? Also, could educational outcomes be affected in any way? Given the linkage between air pollution and education (e.g., Aziz and Elbakidze (2025)), it is entirely possible that the SLF may reduce human capital in the long run.

References

- Aziz, M. and Elbakidze, L. (2025). Invisible barrier: The impact of air quality on chronic school absenteeism in the us. *Environmental and Resource Economics*, 88(3):655–680.
- Bevins, S. N., Pedersen, K., Lutman, M. W., Gidlewski, T., and Deliberto, T. J. (2014). Consequences associated with the recent range expansion of nonnative feral swine. *BioScience*, 64(4):291–299.
- Brown, E. D., Snapp, J. W., Huber, C., Caudill, J., and Grigelis, P. E. (2025). Economic costs of invasive carps in the united states: case study and management implications. *Biological Invasions*, 27(9):191.
- Callaway, B. and Sant’Anna, P. (2021a). did: Difference in differences. R package version 2.1.2.
- Callaway, B. and Sant’Anna, P. (2021b). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230.
- Cary, M. and Orteiz, M. (2026). Gone with the wind: Renewable energy projects and population change. *Energy Economics*, page 109303.
- Cavalier, H., Trasande, L., and Porta, M. (2023). Exposures to pesticides and risk of cancer: Evaluation of recent epidemiological evidence in humans and paths forward. *International journal of cancer*, 152(5):879–912.
- Chalak, M., Polyakov, M., and Pannell, D. J. (2017). Economics of controlling invasive species: a stochastic optimization model for a spatial-dynamic process. *American Journal of Agricultural Economics*, 99(1):123–139.
- Chang, T., Graff Zivin, J., Gross, T., and Neidell, M. (2016). Particulate pollution and the productivity of pear packers. *American Economic Journal: Economic Policy*, 8(3):141–169.

- Chi, G. and Marcouiller, D. W. (2013). Natural amenities and their effects on migration along the urban–rural continuum. *The Annals of Regional Science*, 50(3):861–883.
- Crystal-Ornelas, R. and Lockwood, J. L. (2020). The ‘known unknowns’ of invasive species impact measurement. *Biological Invasions*, 22(4):1513–1525.
- Dara, S. K., Barringer, L., and Arthurs, S. P. (2015). *Lycorma delicatula* (hemiptera: Fulgoridae): a new invasive pest in the united states. *Journal of Integrated Pest Management*, 6(1):20.
- De Bona, S., Barringer, L., Kurtz, P., Losiewicz, J., Parra, G. R., and Helmus, M. R. (2023). lydemapr: an R package to track the spread of the invasive spotted lanternfly (*Lycorma delicatula*, White 1845)(Hemiptera, Fulgoridae) in the United States. *NeoBiota*, 86:151–168.
- Donovan, G. H., Butry, D. T., Michael, Y. L., Prestemon, J. P., Liebhold, A. M., Gatzliolis, D., and Mao, M. Y. (2013). The relationship between trees and human health: evidence from the spread of the emerald ash borer. *American journal of preventive medicine*, 44(2):139–145.
- Donovan, G. H., Michael, Y. L., Gatzliolis, D., Prestemon, J. P., and Whitsel, E. A. (2015). Is tree loss associated with cardiovascular-disease risk in the women’s health initiative? a natural experiment. *Health & place*, 36:1–7.
- Epanchin-Niell, R. S. and Wilen, J. E. (2012). Optimal spatial control of biological invasions. *Journal of Environmental Economics and Management*, 63(2):260–270.
- Epanchin-Niell, R. S. and Wilen, J. E. (2015). Individual and cooperative management of invasive species in human-mediated landscapes. *American Journal of Agricultural Economics*, 97(1):180–198.
- Fan, X., Gómez, M. I., Atallah, S. S., and Conrad, J. M. (2020). A bayesian state-space approach for invasive species management: the case of spotted wing drosophila. *American Journal of Agricultural Economics*, 102(4):1227–1244.

- Gao, X., Song, R., and Timmins, C. (2023). Information, migration, and the value of clean air. *Journal of Development Economics*, 163:103079.
- Gibbons, S., Mourato, S., and Resende, G. M. (2014). The amenity value of english nature: a hedonic price approach. *Environmental and Resource Economics*, 57(2):175–196.
- Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277.
- Hanley, N. and Roberts, M. (2019). The economic benefits of invasive species management. *People and Nature*, 1(2):124–137.
- Harner, A. D., Rowles, T. K., Kar, S., Briggs, L., and Centinari, M. (2025). Feeding by adult spotted lanternfly affects carbon allocation postinfestation in young grapevines. *American Journal of Enology and Viticulture*, 76(1).
- Hilling, C. D., Jiao, Y., Schmitt, J. D., Fabrizio, M. C., Angermeier, P. L., and Orth, D. J. (2025). Management strategy evaluation to assess trade-offs associated with invasive blue catfish fisheries and predation impacts. *Marine and Coastal Fisheries*, 17(1):vtaf004.
- Hilling, C. D., Schmitt, J. D., Jiao, Y., and Orth, D. J. (2023). Predatory impacts of invasive blue catfish in an atlantic coast estuary. *Marine and Coastal Fisheries*, 15(5):e210261.
- Hulme, P. E. (2009). Trade, transport and trouble: managing invasive species pathways in an era of globalization. *Journal of applied ecology*, 46(1):10–18.
- Huron, N. A., Behm, J. E., and Helmus, M. R. (2022). Paninvasion severity assessment of a us grape pest to disrupt the global wine market. *Communications Biology*, 5(1):655.
- Jones, B. A. (2016). Work more and play less? time use impacts of changing ecosystem services: the case of the invasive emerald ash borer. *Ecological Economics*, 124:49–58.

- Jones, B. A. (2017). Invasive species impacts on human well-being using the life satisfaction index. *Ecological Economics*, 134:250–257.
- Jones, B. A. (2020a). Invasive species control, agricultural pesticide use, and infant health outcomes. *Land Economics*, 96(2):149–170.
- Jones, B. A. (2020b). Labor market impacts of deforestation caused by invasive species spread. *Environmental and Resource Economics*, 77(1):159–190.
- Jones, B. A. (2023). Can invasive species lead to sedentary behavior? the time use and obesity impacts of a forest-attacking pest. *Journal of Environmental Economics and Management*, 119:102800.
- Jones, B. A. and McDermott, S. M. (2018). Health impacts of invasive species through an altered natural environment: assessing air pollution sinks as a causal pathway. *Environmental and Resource Economics*, 71(1):23–43.
- Kim, H. N., Boxall, P. C., and Adamowicz, W. (2016). The demonstration and capture of the value of an ecosystem service: A quasi-experimental hedonic property analysis. *American Journal of Agricultural Economics*, 98(3):819–837.
- Kleespies, M. W., Dörge, D. D., Peter, N., Schantz, A. V., Skaljic, A., Feucht, V., Burger-Schulz, A. L., Dierkes, P. W., and Klimpel, S. (2024). Identifying opportunities for invasive species management: an empirical study of stakeholder perceptions and interest in invasive species. *Biological invasions*, 26(8):2561–2577.
- Ladin, Z. S., Eggen, D. A., Trammell, T. L., and D’Amico, V. (2023). Human-mediated dispersal drives the spread of the spotted lanternfly (*lycorma delicatula*). *Scientific Reports*, 13(1):1098.
- Macpherson, M. F., Kleczkowski, A., Healey, J. R., and Hanley, N. (2018). The effects of disease

- on optimal forest rotation: a generalisable analytical framework. *Environmental and Resource Economics*, 70(3):565–588.
- Multi-Resolution Land Characteristics Consortium (2024). National land cover database (nlcd) 2021 products. Available from the Multi-Resolution Land Characteristics Consortium.
- Pimentel, D., Zuniga, R., and Morrison, D. (2005). Update on the environmental and economic costs associated with alien-invasive species in the united states. *Ecological economics*, 52(3):273–288.
- Richards, T. J., Shanafelt, D. W., and Fenichel, E. P. (2014). Foreclosures and invasive insect spread: the case of asian citrus psyllid. *American Journal of Agricultural Economics*, 96(3):615–630.
- Roback, J. (1982). Wages, rents, and the quality of life. *Journal of Political Economy*, 90(6):1257–1278.
- Sanda, M. and Lindeman, S. (2025). County-level eastern u.s. forest loss since 2010 [dataset]. <https://doi.org/10.5061/dryad.r4xgxd2s7>.
- Sharp, R. L., Larson, L. R., and Green, G. T. (2011). Factors influencing public preferences for invasive alien species management. *Biological Conservation*, 144(8):2097–2104.
- Strömbom, D., Sands, A., Graham, J. M., Crocker, A., Cloud, C., Tulevech, G., and Ward, K. (2024). Modeling human activity-related spread of the spotted lanternfly (*lycorma delicatula*) in the us. *Plos One*, 19(8):e0307754.
- Sun, J., Koski, T.-M., Wickham, J. D., Baranchikov, Y. N., and Bushley, K. E. (2024). Emerald ash borer management and research: decades of damage and still expanding. *Annual Review of Entomology*, 69(1):239–258.

Urban, J. M., Calvin, D., and Hills-Stevenson, J. (2021). Early response (2018–2020) to the threat of spotted lanternfly, *lycorma delicatula* (hemiptera: Fulgoridae) in pennsylvania. *Annals of the Entomological Society of America*, 114(6):709–718.

US National Park Service (2020). Invasive & Non-Native Species. <https://www.nps.gov/subjects/invasive/learn.htm>.

Wolf, D., Gopalakrishnan, S., and Klaiber, H. A. (2022). Staying afloat: The effect of algae contamination on lake erie housing prices. *American Journal of Agricultural Economics*, 104(5):1701–1723.